



Vinod Kumar Neelakantam

EMBEDDED PLATFORM INTEGRATION

Links

 LinkedIn:

linkedin.com/in/vinodneelakantam

 GitHub:

github.com/vinodneelakantam

Skills

Build & Release Automation

C/C++, Python, Groovy, Shell

Methodologies

Automotive SPICE (ASPICE) SWE.5
Agile, CI/CD/CT, TDD, SW/HW
Debugging

Automotive Tools

Development/Integration

DevOps for Embedded

CMake, Jenkins (CloudBees),
Docker/Kubernetes, Artifactory,
Grafana/Elastic Search

CI/CD Quality & Security

SAST/SCA Checks, SBOM
(SPDX/CycloneDX)

C/C++ Build Systems

Yocto, Zephyr, AOSP, Custom

Automotive Cybersecurity

TARA, ISO/SAE 21434, CSMS
(UNECE R155), SUMS (UNECE R156)

Secure OTA/SOTA

Package Signing, Integrity Checks,
Partial Update Strategy

Artifact & Dependency

Assurance, CVE Triage

Protocols

UART, I2C, SPI, CAN, Ethernet

Contact

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Personal Information

- **Nationality:** Indian
- **Work Authorization:** Germany Permanent Residence Permit (Niederlassungserlaubnis)
- **Languages:** English(C1), German(A1)
- **Driving License:** Class B
- **Education:** Master's in Embedded Systems, India, Hyderabad, 2013

About Me

- Embedded Platform Integration Engineer with 12+ years in automotive software, delivering robust ECU and ADAS integration across device drivers, board bring-up, system validation, and platform release management.
- Designs and operates ASPICE-aligned CI/CD and release ecosystems that increase delivery reliability, reduce integration risk, and improve traceability for distributed engineering teams.
- Combines deep embedded build expertise (C/C++, Yocto, AOSP, Zephyr, Bricks-Evo) with practical DevOps execution to support multi-variant products at production scale.
- Focused for 7+ years on automotive cybersecurity integration, including TARA-based engineering, ISO/SAE 21434 work products, and secure OTA/SOTA delivery with signing and integrity controls.
- Brings end-to-end PDLC ownership, strong stakeholder alignment, and hands-on problem solving across development, integration, validation, release, and audit readiness.

Core Competencies

- Automotive Embedded Integration: ADAS and ECU platform integration, multi-variant delivery, build and dependency governance.
- CI/CD and DevSecOps: Pipeline architecture, quality gates, artifact traceability, SBOM-aware release workflows.
- Automotive Cybersecurity: TARA support, ISO/SAE 21434 alignment, CSMS (UNECE R155) and SUMS (UNECE R156) evidence readiness.
- Process and Compliance: ASPICE SWE.5-aligned integration practices, release documentation, and audit preparation.

Certifications

- ISTQB Certified Tester - Foundation Level (CTFL)

Current Role Focus

- Embedded Platform Integration Engineer - Automotive Software Platforms
- CI/CD and Release Integration for ADAS and Embedded ECU Programs
- Automotive Cybersecurity Integration and Secure OTA/SOTA Delivery

Work Experience

Embedded Platform Integration

AUMOVIO (CONTINENTAL) — Ulm

09/2023 - Present

- Led ADAS platform integration and multi-variant C/C++ builds on Continental Bricks-Evo, including dependency and configuration governance.
- Built and maintained CI/CD pipelines (Jenkins, CloudBees, Docker, Kubernetes, Artifactory) with automation tooling in Python.
- Implemented secure OTA/SOTA packaging (PDX, signing, partial updates) and release traceability aligned to ISO/SAE 21434 expectations.
- Contributed to TARA deliverables and CSMS/SUMS evidence readiness (UNECE R155/R156) through controlled release workflows.
- Introduced security and quality gates (SAST/SCA, dependency risk checks, integrity validation) across embedded pipelines.
- Improved delivery reliability through pipeline optimization, release lifecycle ownership, and L2/L3 build infrastructure support.

Solution Integrator (ADAS, Surround View systems)

AUMOVIO (CONTINENTAL) — Ulm

09/2021 - 08/2023

- Coordinated ADAS release planning and cross-functional integration with project managers, architects, test teams, and customer milestones.
- Designed stable CI/CD workflows with Jenkins, CMake, GitHub, and Artifactory for repeatable builds and delivery control.
- Implemented ASPICE SWE.5-compliant integration and release processes, including strategy, documentation, and quality checkpoints.
- Established automated smoke/regression validation with HIL/SIL execution and early defect detection.
- Performed system-level debugging and protocol validation (CAN, Ethernet, SPI, I2C, UART) for camera/surround-view programs.

Software Integrator (ADAS, Camera Mirror Replacement)

SMR AUTOMOTIVE GLOBAL — Stuttgart

07/2017 - 08/2019

- Optimized Jenkins-based CI/CD for camera mirror ECU programs, improving build stability and release efficiency.
- Defined ASPICE-compliant integration/test strategies and maintained audit-ready process documentation.
- Integrated VectorCAST and PRQA into CI for automated validation and static quality controls.
- Implemented smoke/regression/timing/data-flow test workflows aligned with quality targets.
- Led audit preparation and process improvement activities, supporting successful customer delivery assessments.

Release Candidate Engineer (ARC products)

SYNOPSYS INC — Hyderabad

08/2016 - 07/2017

- Validated Synopsys ARC release candidates across embedded software, operating systems, multimedia stacks, and FPGA development boards.
- Executed interoperability testing to protect product dependencies and release quality.

Software Integrator (Radars)

INFOSYS — Bengaluru

03/2014 - 08/2016

- Delivered automotive software integration for radar programs in collaboration with global customer teams.
- Led UDS fault-management integration for Hyundai-Kia short-range side radar and improved diagnostic behavior.

Software Integrator (Engine Management Systems)

ROBERT BOSCH ENGINEERING — Coimbatore Area

06/2012 - 08/2013

- Performed unit/integration validation for engine management software, including boundary testing and resource measurement.
- Automated smoke tests on ETAS LAB Car and supported ASPICE quality coordination activities.